Individual and Common Attack: Enhancing Transferability in VLP Models Through Modal Feature Exploitation

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***Abstract*—Vision–Language Pretrained (VLP) models exhibit strong multimodal understanding and reasoning capabilities, finding wide application in tasks such as image–text retrieval and visual grounding. However, they remain highly vulnerable to adversarial attacks, posing serious reliability concerns in safety-critical scenarios. We observe that existing adversarial examples optimization methods typically rely on individual features from the other modality as guidance, causing the crafted adversarial examples to overfit that modality's learning preferences and thus limiting their transferability. In order to further enhance the transferability of adversarial examples, we propose a novel adversarial attack framework, I&CA (Individual & Common feature Attack), which simultaneously considers individual features within each modality and common features cross-modal interactions. Concretely, I&CA first drives divergence among individual features within each modality to disrupt single-modality learning, and then suppresses the expression of common features during cross-modal interactions, thereby undermining the robustness of the fusion mechanism. In addition, to prevent adversarial perturbations from overfitting to the learning bias of the other modality, which may distort the representation of common features, we simultaneously introduce augmentation strategies to both modalities. Across various experimental settings and widely recognized multimodal benchmarks, the I&CA framework achieves an average transferability improvement of 6.15% over the state-of-the-art DRA method, delivering significant performance gains in both cross-model and cross-task attack scenarios.**

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I. Introduction

VISION-LANGUAGE Pretrained (VLP) models are widely applied in multimodal tasks [1], [2], such as image captioning, visual question answering, and cross-modal retrieval. However, they are vulnerable to adversarial attacks, as shown in recent works [3], [4], [5], [6], [7], [8].

Compared to backdoor attacks [6], malicious triggers are usually activated only under specific conditions. This type of attack relies on the model's behavior in particular contexts and therefore cannot be applied in a wide range of scenarios. In contrast, misleading the model by adding imperceptible perturbations to the input has broader applicability, as it can instantly affect the model during the inference phase without relying on specific trigger conditions [3], [4], [5]. Adversarial attacks based on input perturbations can be categorized into two primary types: white-box. White-box attacks usually require obtaining internal information, such as the model's structure and parameters, whereas do not have such constraints. For example, Co-Attack [3] is a typical white-box attack on VLP models. However, white-box attacks are limited in their applicability in the real world. Therefore, the current research focus has been especially transfer-based adversarial attacks [9], [10].

The so-called transfer attack means generating adversarial examples on a local *surrogate model* to attack an unknown *target model*. In other words, it uses the surrogate model to simulate the target model. Lu et al. [4] proposed the SGA transfer-based attack. This approach enhances the dissimilarity between image-text pairs at the ensemble level to generate transferable adversarial examples. However, the generation of adversarial examples heavily depends on examples along the adversarial trajectory, leading to overfitting the surrogate model. This overfitting will reduce the cross-model attack ability of adversarial examples, that is, their *transferability*. To reduce the risk of overfitting, Gao et al. [5] proposed diversification along the intersection region of the adversarial trajectory (named DRA) based on the SGA framework.

The limitations of existing attack methods (such as SGA [4] and DRA [5]) stem from their excessive reliance on unimodal features. Specifically, during the generation of adversarial examples, the perturbation update of one modality (e.g., text) is typically strongly guided by the individual features of

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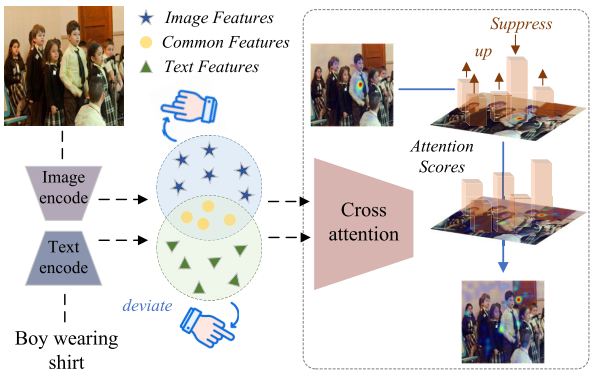

Fig. 1. Illustration of our proposed I&CA's main idea. Deviate from individual features and suppress common features.

another modality (e.g., image), without explicitly considering the stable common features across different models. *Individual features* refer to modality-specific information, such as syntactic structure and word order in text, or local texture and pixel distribution in images. *Common features* represent the semantic consistency between text and image modalities and are the information that different VLP models commonly rely on in multimodal task decision-making [11]. Therefore, over-reliance on the individual features of another modality leads to perturbations being optimized to overfit the learned biases of that modality's model, thereby limiting the transferability of adversarial examples. Mechanistically, since the gradients are computed solely based on the modality-specific features of the source model, the resulting perturbations are optimized to exploit the model-specific biases and feature preferences. Different VLP models often implement distinct architectures or focus on different feature granularities (e.g., ViT vs. CNN for images, or attention patterns for text). Therefore, adversarial perturbations that are highly effective on the surrogate model fail to generalize to the target model, leading to insufficient transferability.

To this end, we propose the individual & Common Attack (I&CA), which clearly distinguishes the handling of individual features and common features. Individual loss guides the perturbations to diversify intra-modal features without relying on the surrogate model's preferences, while common loss suppresses the model's attention to shared cross-modal features. By perturbing both the modality-specific and modality-shared subspaces, the adversarial examples generated by I&CA avoid overfitting to any single surrogate model and better exploit the features that are consistently used across different VLP architectures, as shown in Fig. 1. Additionally, we apply dual-modal augmentation to prevent adversarial examples from overfitting the learning preferences of the surrogate model, thereby weakening the representation of common features under modality interactions. Specifically, in the text modality, we combine adversarial texts with original texts to construct an augmented text set. For the image modality, we apply a dual-domain scale transformation (spatial and frequency domains). Finally, to evaluate the effectiveness of the attack framework, we use ALBEF [12] and TCL [13] as surrogate models and test the attack performance of the adversarial examples on several widely used multimodal datasets (such as Flickr30K [14], MSCOCO [15], and RefCOCO [16]).

Our contributions can be summarized as follows:

- We proposed a novel adversarial attack framework I&CA, which guides the deviation of individual features within modalities while suppressing common features under modality interactions. By guiding individuality features to diverge while suppressing common features, the framework reduces overfitting to the surrogate model, thereby enhancing the transferability of adversarial examples across different models and tasks.
- We proposed dual-modal augmentation to prevent the individual feature of adversarial examples from overfitting the learning preferences of the surrogate model, which would otherwise weaken the expression of common features. For the discrete nature of the text, we adopt a text expansion strategy, while for the image, we employ a dual-domain scale transformation.
- Experimental results show that our method exhibits good transferability (average improvement of **6.15%** compared to the current DRA attack method) across models and demonstrates exceptional attack performance in cross-task applications.

II. Related Work

This section provides an overview of the overall architecture of VLP models, representative downstream tasks, and current adversarial attack methods based on pretrained vision-language models.

A. Vision-Language Pretraining Models

VLP models learn the semantic alignment between different modalities for multimodal task processing by pretraining on large-scale image-text pairs. Based on how VLP models extract features, they can be divided into two main architectures [17]: single-stream architecture and dual-stream architecture. The single-stream architecture concatenates textual and visual features together and inputs them into a single Transformer module, utilizing a combined attention mechanism to fuse multimodal inputs (*e.g.*, ALBEF [12], TCL [13]). The dual-stream architecture, on the other hand, does not concatenate textual and visual features; instead, it uses independent Transformers for feature extraction and then integrates cross-modal information through a cross-attention mechanism (*e.g.*, CLIP [18]).

B. VLP's Downstream Tasks

VLP models are widely applied to various downstream vision-language tasks.

- Image-Text Retrieval (**ITR**) [19], [20] aims to retrieve relevant data from one modality based on a query from the other. It is typically divided into two sub-tasks: Text Retrieval (TR), which involves retrieving matching text given an image query, and Image Retrieval (IR), where the goal is to find relevant images based on a text query;
- Visual Grounding (**VG**) [21], [22], [23] focuses on accurately locating image regions corresponding to entities or concepts described in natural language;

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- Image Captioning (**IC**) [24], [25], [26], which involves generating a logically coherent and semantically accurate textual description of a given visual input.

C. Adversarial Attacks

Adversarial attacks can generally be classified into two categories: white-box attacks [3], [27], [28], [29], [30], [4], [5], [31], [32]. White-box attacks refer to situations where the attacker has full access to the target model's internal information, including its structure, parameters, etc., while occur when the attacker does not have direct access to the target model's internal information and can only interact with the model through input-output queries. Specifically, can be divided into query-based attacks, such as VLattack [33], and transfer-based attacks, such as SGA [4] and DRA [5]. However, query-based attacks rely on the outputs of the target network, so exploring transfer-based attacks is more practical.Transferable adversarial attacks refer to adversarial examples generated in a white-box setting (*i.e.*, using a surrogate model) that can successfully induce a black-box model (*i.e.*, target model) to produce incorrect outputs. To generate more realistic adversarial examples, this paper primarily investigates the transferability of adversarial examples in a black-box setting.

In early multimodal research, researchers primarily attacked multimodal models by perturbing individual modalities (*e.g.*, Sep-attack [3]). Due to the lack of sufficient utilization of cross-modal interactions, the effectiveness was limited. Zhang et al. [3] first explored the adversarial robustness of VLP models in a white-box setting. Lu et al. [4] proposed the integrated-level text-image similarity (SGA), which was the first to study the transferability of adversarial examples on VLP models. Gao et al. [5] optimized the path to the generation of adversarial examples based on SGA, further improving the transferability of adversarial examples. However, existing methods primarily rely on the guidance of individual features from the other modality while overlooking the role of common features under modality interactions.

III. Method

This section primarily explains the design motivation and core ideas of the I&CA attack framework, detailing how to achieve divergence of individual features while suppressing common features in cross-modal interactions. It also clarifies the necessity and role of introducing modality augmentation strategies. Finally, it presents the overall process of the attack framework and the algorithm implementation.

A. Motivation

In adversarial attacks on VLP models, the attacker aims to induce semantic deviations between the images and texts while limiting the perturbation magnitude of the input images and texts, causing the model to output incorrect results. Specifically, a clean image-text pair $(v,\ t)$ is mapped to embedding representations e_v and e_t by the image encoder and text encoder, respectively. The attacker introduces image perturbation δ_v and text perturbation δ_t to generate adversarial examples.

$$\arg\max_{\delta_v,\delta_t}\ \mathcal{L}(e'_v, e'_t),$$
$$\text{s.t.}\quad \|\delta_v\|_p \le \epsilon_v,\quad \|\delta_t\|_p \le \epsilon_t, \tag{1}$$

where e'_v and e'_t represent the perturbed image embedding and text embedding, and δ_v and δ_t denote the magnitudes of perturbations ϵ_v and ϵ_t for the images and texts, respectively. $\mathcal{L}$ denotes the function used to measure the differences between embeddings. In black-box settings, we want to maximize the function $\mathcal{L}$ of the target model, not only the local full-access model. In other words, we hope to improve the transferability that crosses multiple models.

Existing studies have shown that cross-modal interactions and data diversity help improve the transferability of adversarial examples [27], [33], [34]. However, current methods primarily rely on the guidance of individual features to achieve cross-modal interactions, neglecting common features [4], [5]. Additionally, current modality augmentation mainly focuses on the image modality while overlooking the text modality. Therefore, based on the limitations of current research, we propose to enhance transferability by using individual loss to guide the divergence of individual features within modality while employing common loss to suppress common features under modality interactions. Furthermore, to prevent adversarial examples from overfitting the learning preferences of the surrogate model [35], thereby weakening the expression of common features, we apply data augmentation to both modalities.

B. Individual & Common Loss

In multimodal tasks, the individual features within each modality largely determine the model's understanding of each modality. Recall that individual features refer to the semantic information independently expressed in the image and text modalities. Before modality interactions, we leverage cross-modal data guidance to disrupt the alignment between modalities, achieving mutual deviation of individual features and further enhancing the independence of individual features within each modality. Specifically, our loss function is defined as follows:

$$\mathcal{L}_{\text{Indiv}} = \frac{f_v(v') \cdot f_t(t')}{\|f_v(v')\|\|f_t(t')\|}, \tag{2}$$

where $f_v(v')$ and $f_t(t')$ represent the feature representations of the image and text modalities, respectively. By minimizing the individual loss, the mutual deviation of individual features within each modality is achieved.

During modality interaction, common features of the modalities play a key role in the decision-making process of multimodal models [11]. The cross-attention module in VLP models, as the core mechanism for achieving modality interaction [17], is closely related to the perception of modality common features. The attention scores in the cross-attention mechanism reflect the contribution of common features in the image-text interaction to the final decision when the model processes image-text inputs, and their concrete manifestation in Fig. 3. Therefore, these scores indicate how the model focuses on different parts of the input data [36]. A large variance in attention scores means that the model's attention to common features in modality interaction varies significantly. Based on this, we propose a perturbation generation method

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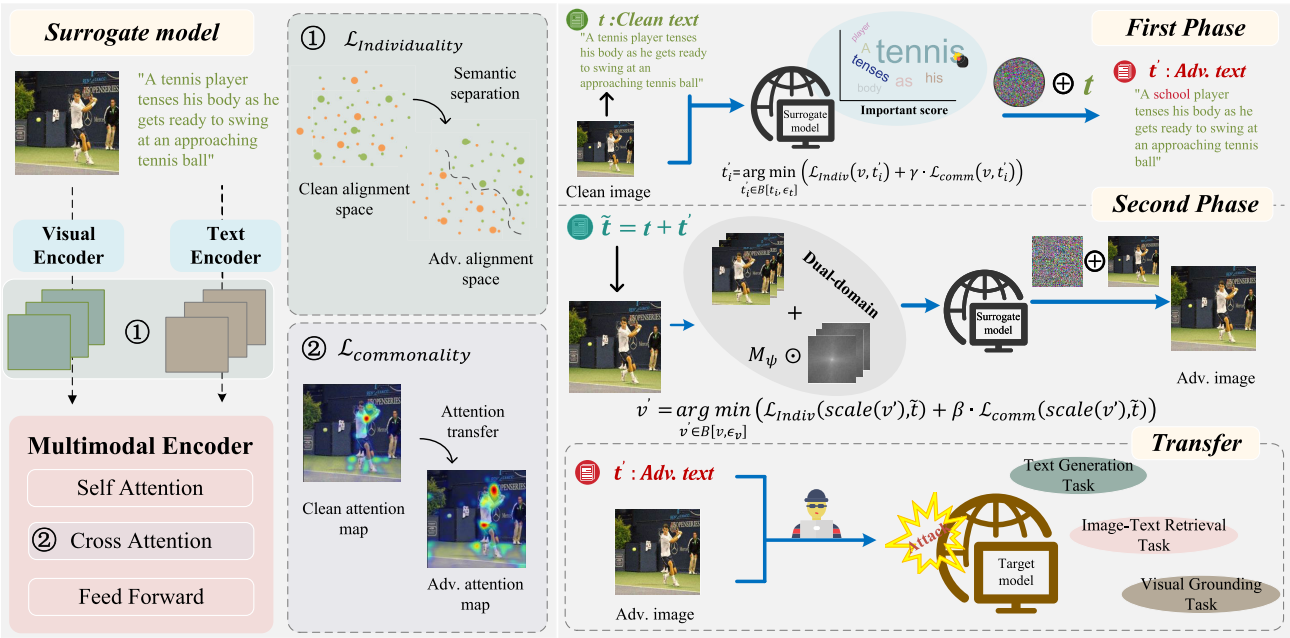

Fig. 2. Illustration of I&CA framework. The left side introduces the individual loss and common loss, while the right side illustrates our overall training process.

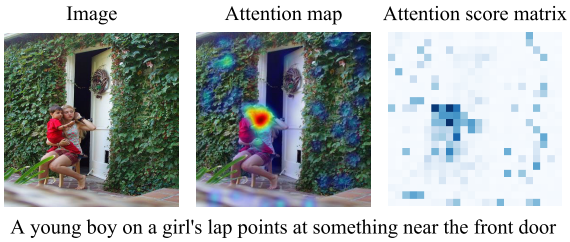

Fig. 3. The manifestation of common features. In the attention score matrix, darker colors indicate higher attention scores.For example, the areas with darker colors in the attention score matrix correspond to the text tokens "boy" and "points".

based on the cross-attention mechanism. This method minimizes the attention score matrix to reduce the differences in the model's attention to common features and disrupt the original feature attention regions. Specifically, we extract the attention score matrices from each layer and average the scores across layers to generate a matrix that integrates multi-layer information, which we use as the representation of the common features. Finally, we minimize this score matrix to disperse the model's attention across specific regions, thereby disrupting the model's learning of common features. The definition of common loss is as follows:

$$\mathcal{L}_{\text{comm}} = \text{Var}\left(\frac{1}{N}\sum_{i=1}^{N} \text{S}_{\text{atten}}(v', t')_i\right) \tag{3}$$

$$\text{S}_{\text{atten}}(v', t')_i = \text{softmax}\left(\frac{(v' W_i^Q)(t' W_i^K)^\top}{\sqrt{d_k}}\right) \tag{4}$$

where $S_{atten}(\cdot)_i$ represents the result of the attention score after applying softmax in the i-th layer. W_i^Q, W_i^K are the learnable projection matrices corresponding to the i-th attention head. Var$(\cdot)$ computes the input variance. By minimizing the common loss, we suppress the activation of common features, facilitating attention transfer.

C. Dual-Modal Augmentation

Since individual features overfit the learning preferences of the surrogate model [35], further affecting the expression of common features, modality augmentation becomes particularly important to prevent overfitting and ensure the effective expression of common features. It is worth noting that we primarily adopt existing data augmentation methods to implement dual-modal augmentation, as the specific augmentation techniques are not the core innovation of this study.

1) Text Modality Augmentation: In our research method, to avoid adversarial texts that overfit the learning preferences of the surrogate model, thus leading to weakened representation of common features and reduced guidance for the image modality. We treat the set of original and adversarial texts as a means of data augmentation for the text modality. Due to the discreteness of the text modality, modifications are mainly focused on word-level changes [37], [38] and sentence-level variations [39]. Specifically, using the original images v as guidance, adversarial texts t' are generated through BERT-ATTACK [40], which is then combined with the original texts to form an augmented text set $\tilde{t} = \{t_1, t_2, \ldots, t_m; t_1', t_2', \ldots, t_M'\}$.

2) Image Modality Augmentation: To more effectively capture the common features involved in modality interactions, we also need to enhance the image modality. The image modality typically exists in two forms: the spatial domain and the frequency domain. Methods based on spatial domain input transformations have been widely studied, including input diversification methods (DIM) [41], translation-invariance methods (TIM) [42], scale-invariance methods (SIM) [43], and mixed image iteration methods (Admix) [44]. These methods primarily enhance the transferability of adversarial examples by altering the spatial distribution characteristics of images, such as through translation, scaling, or pixel arrangement diversification. However, spatial domain input transformations mainly focus on the geometric features of images in spatial

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positions, failing to capture local variations within images [45]. In contrast, in frequency domain input transformations, high-frequency information often reflects local detail information in the pixel space [9], [10], [46]. Therefore, in our study, to fully explore the expression of common features in interaction with the text modality, we perform dual-domain scale transformation on the image modality. Moreover, existing research has shown that dual-domain interaction can promote information flow and the learning of complementary representations [47].

Specifically, in the spatial domain, we use the SIM [43] based on deep learning models to construct multi-scale images $v_s = \{s_1, s_2, \ldots, s_m\}$, capturing diverse spatial geometric features. In the frequency domain, we apply the Discrete Cosine Transform (DCT) to convert the images v from the spatial domain to the frequency domain, obtaining their frequency spectrum. Next, the spectrum is multiplied by a random scaling factor $M_\psi \in (1-\rho, 1+\rho)$, where ρ is the scaling factor that controls the scaling in the frequency domain. Finally, the Inverse Discrete Cosine Transform (IDCT) is used to convert the frequency domain signal back to the spatial domain, yielding the transformed image example v_τ. The defined frequency domain scaling is as follows:

$$v_\tau = \mathcal{D}^{-1}\big(\mathcal{D}(v) \odot M_\psi\big), \tag{5}$$

where $\mathcal{D}$ and $\mathcal{D}^{-1}$ are DCT and IDCT, respectively, and $\odot$ is the element-wise multiplication. Ultimately, a multi-scale transformed image set is generated $\tilde{v} = \{v_s; v_\tau\}$. The expanded input set $(\tilde{t}, \tilde{v})$ is used to generate adversarial data v'. The detailed experimental process is elaborated in Section III-D.

Based on the previous sections, we summarize the I&CA attack framework as follows: Before modality interactions, individual loss is used to guide the deviation of individual features; During modality interactions, common loss is applied to suppress cross-modal interactions, reducing the model's reliance on key regions. Additionally, to prevent individual features from overfitting to model preferences during optimization, thereby weakening the expression of common features, we introduce dual-modal augmentation.

D. Perturbation Optimization

In the previous sections, we provided a detailed explanation of the motivation behind our method and the design of the loss functions. In this section, we will elaborate on the overall procedure of our approach.

In the first phase, we use the original images v as text guidance to generate adversarial texts t', and the optimization is as follows:

$$t'_i = *\arg\min_{t'_i \in B[t_i, \epsilon_t]} \big(\mathcal{L}_{\text{Indiv}}(v, t'_i) + \gamma \cdot \mathcal{L}_{\text{comm}}(v, t'_i)\big). \tag{6}$$

In the second phase, the obtained t'_i are combined with t_i to form an enhanced text set $\tilde{t}$ as image v guidance. v' are mainly updated using the following approach:

$$v' = *\arg\min_{v' \in B[v, \epsilon_v]} \big(\mathcal{L}_{\text{Indiv}}(scale(v'), \tilde{t}) + \beta \cdot \mathcal{L}_{\text{comm}}(scale(v'), \tilde{t})\big), \tag{7}$$

where $scale(\cdot)$ represents dual-domain scale transformation. We use $B[v, \epsilon_v]$ and $B[t, \epsilon_t]$ to represent the searching spaces for optimizing adversarial image and text. ϵ_v denotes the maximal perturbation bound for the image, and ϵ_t denotes the maximal number of changeable words in the caption. Our overall process is shown in Fig. 2 and Algorithm 1.

Algorithm 1 **I**ndividual & **C**ommon **A**ttack

Input: The clean image-text pair (v, t), interation steps K, number of paired caption M, max perturbation ϵ_v and ϵ_t
Parameter: Weight parameters γ and β, spatial domain scaling transformation $S = \{0.50, 0.75, 1.00, 1.25, 1, 50\}$ and frequency domain scaling intensity $\rho \in (0, 1)$.
Output: Adversarial image v' and adversarial text t'
// *phase I*: Forming an augmented text collection along with the original text $\tilde{t}$;

```
for i = 1 to M do
    t'_i ← argmin_{t'_i ∈ B[t_i, ε_t]} (L_Indiv(v, t'_i) + γ · L_comm(v, t'_i))
end for
t̃ = {t_1, t_2, ..., t_m; t'_1, t'_2, ..., t'_M}
```

// *phase II*: Build adversarial image v' using $\tilde{t}$ in phase I;

```
for j = 1 to N do
    ṽ_j = {v_s(j), v_τ(j)}
    v'_{j+1} ← argmin_{v' ∈ B[v, ε_v]} (L_Indiv(ṽ_j, t̃) + β L_Comm(ṽ_j, t̃))
end for
return t'_i, v'
```

IV. Experiments

This section primarily presents our experimental setup, provides a detailed evaluation and presentation of the results, and conducts an ablation study on key components.

A. Experimental Setup

1) Datasets: In this study, we used widely recognized multimodal datasets, including Flickr30K [14], MSCOCO [15], and RefCOCO+ [16]. Flickr30K is commonly used for ITR tasks and consists of 31,783 images, each paired with five descriptive texts. MSCOCO is typically used for ITR and IC tasks, containing 123,287 images, each paired with approximately five descriptive captions. RefCOCO+ is mainly utilized for VG tasks and includes 19,992 images from the MSCOCO training set and 141,564 referring expressions. The primary purpose of RefCOCO+ is to evaluate the model's ability to locate target objects through natural language captions.

2) Model Architectures: We adopted two standard VLP model architectures: single-stream and dual-stream. Since the dual-stream architecture mainly focuses on intra-modal feature learning and lacks cross-modal interaction mechanisms, we selected ALBEF [12] and TCL [13], both based on the single-stream architecture, as surrogate models in our attack framework. Although TCL and ALBEF share the same architecture, they differ in their pretraining objectives and areas of focus. To demonstrate that the I&CA attack framework is also effective against dual-stream models, we chose CLIP as the target model. Notably, CLIP [18] uses two different image feature extraction modules: ViT-B/16 [48] (CLIP_{vit}) and ResNet-101 [49] (CLIP_{cnn}).

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TABLE I

THE ATTACK SUCCESS RATES (ASR%) OF THE BASELINE METHODS ON THE ITR TASK OF THE FLICKR30K AND MSCOCO DATASETS, USING ALBEF AND TCL AS THE SURROGATE MODEL. THE GRAY SHADING INDICATES WHITE-BOX SETTINGS, WHILE BOLD NUMBERS REPRESENT THE BEST RESULTS

Flickr30K (Image-Text Retrieval)													
Model	Method	ALBEF			TCL			CLIP_{vit}			CLIP_{cnn}		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
ALBEF	Co-Attack	97.08	94.59	92.60	39.52	20.40	14.53	29.82	11.73	6.61	31.29	11.73	5.77
	SGA	**100.00**	99.70	99.70	87.57	77.69	72.44	36.93	18.90	13.01	41.00	21.88	15.76
	DRA	99.79	99.80	99.80	90.62	81.11	75.55	45.40	24.51	17.78	50.45	29.07	21.01
	Ours	**100.00**	**100.00**	**100.00**	**97.79**	**93.97**	**92.08**	**51.66**	**29.70**	**22.15**	**55.68**	**34.78**	**25.85**
TCL	Co-Attack	49.84	27.45	60.63	91.68	85.23	80.96	32.64	13.40	6.81	32.06	14.27	8.14
	SGA	93.12	86.97	84.00	100.00	100.00	100.00	41.96	24.40	16.77	50.70	29.81	20.29
	DRA	94.68	90.48	87.90	100.00	100.00	99.90	46.85	26.34	18.50	51.11	29.89	21.43
	Ours	**97.81**	**94.59**	**93.50**	**100.00**	**100.00**	**100.00**	**49.08**	**28.66**	**20.63**	**54.53**	**34.99**	**25.85**

Flickr30K (Text-Image Retrieval)													
Model	Method	ALBEF			TCL			CLIP_{vit}			CLIP_{cnn}		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
ALBEF	Co-Attack	98.36	96.41	94.86	51.24	31.90	24.41	38.92	23.31	17.01	41.99	25.18	18.55
	SGA	99.93	99.92	99.90	87.64	77.47	71.61	46.97	28.75	22.20	50.43	32.78	24.76
	DRA	99.93	99.88	99.84	91.00	81.99	77.34	56.48	38.05	29.18	59.42	41.99	33.84
	Ours	**100.00**	**100.00**	**99.98**	**97.36**	**93.55**	**91.04**	**62.66**	**43.84**	**35.61**	**66.21**	**48.28**	**40.14**
TCL	Co-Attack	60.63	41.59	33.26	95.48	90.32	86.74	42.69	26.44	20.37	47.82	30.47	23.13
	SGA	92.52	86.26	82.39	**100.00**	99.98	99.98	49.90	32.96	25.67	55.99	37.94	29.07
	DRA	94.95	90.20	87.63	**100.00**	99.98	99.96	56.84	38.95	30.99	62.11	43.76	35.41
	Ours	**97.26**	**94.69**	**93.09**	**100.00**	**99.98**	**99.98**	**58.96**	**40.71**	**33.34**	**64.87**	**46.12**	**37.20**

MSCOCO (Image-Text Retrieval)													
Model	Method	ALBEF			TCL			CLIP_{vit}			CLIP_{cnn}		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
ALBEF	Co-attack	96.65	94.74	93.12	57.33	37.24	28.53	50.71	33.10	26.44	52.06	33.89	27.47
	SGA	99.90	99.75	99.69	87.04	75.98	68.82	63.91	46.81	38.59	63.91	47.6	40.33
	DRA	99.97	99.72	99.51	88.76	78.44	71.45	69.63	52.41	43.92	69.06	51.97	44.02
	Ours	**99.97**	**99.89**	**99.86**	**96.51**	**92.00**	**88.93**	**76.15**	**61.54**	**52.03**	**74.38**	**58.04**	**49.31**
TCL	Co-Attack	65.22	45.39	36.43	94.95	91.18	89.08	55.28	38.04	29.73	56.68	37.31	29.82
	SGA	91.70	85.81	82.05	**100.00**	99.98	99.98	65.51	48.93	41.48	65.39	49.52	42.73
	DRA	94.90	89.82	86.62	**100.00**	99.98	99.98	69.84	54.12	45.81	69.84	52.47	45.47
	Ours	**97.83**	**95.80**	**94.32**	**100.00**	**99.99**	**99.99**	**72.83**	**58.11**	**49.68**	**71.52**	**55.02**	**47.56**

MSCOCO (Text-Image Retrieval)													
Model	Method	ALBEF			TCL			CLIP_{vit}			CLIP_{cnn}		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
ALBEF	Co-attack	98.33	96.60	95.3	64.19	46.17	37.83	57.36	42.19	35.53	60.74	45.90	38.77
	SGA	99.93	99.84	99.74	87.97	77.27	70.57	69.57	55.33	47.65	70.66	55.88	48.89
	DRA	99.93	99.78	99.67	89.79	80.03	73.63	75.12	61.09	53.74	75.34	61.51	54.30
	Ours	**99.97**	**99.93**	**99.89**	**96.40**	**91.80**	**88.53**	**79.91**	**67.26**	**60.49**	**79.47**	**67.14**	**59.55**
TCL	Co-Attack	72.41	56.37	48.16	97.87	95.32	93.42	62.33	46.90	39.68	66.45	49.95	42.72
	SGA	92.41	86.46	82.72	**100.00**	99.99	99.98	68.51	54.66	47.49	70.68	57.64	50.02
	DRA	95.3	91.30	88.45	**100.00**	99.99	99.98	74.25	60.95	53.93	75.56	63.69	55.34
	Ours	**97.98**	**95.83**	**94.41**	**100.00**	**99.99**	**99.99**	**76.90**	**63.82**	**56.87**	**77.25**	**68.83**	**57.92**

3) Adversarial Attack Settings: We adopted different strategies for texts and images in our method. For texts, we used the concept of BERT-Attack [40] to attack texts with a perturbation range of $\epsilon_t = 1$ and a word list length $W = 10$. For images, we used PGD [50] to generate adversarial examples with a perturbation range of $\epsilon_v = 8/255$, $\alpha = 2/255$,

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TABLE II
HARDWARE CONFIGURATIONS AND SOFTWARE CONFIGURATIONS FOR ALL REPORTED RESULTS

Hardware Configurations	
GPU(s)	NVIDIA RTX A6000 GPU
GPU Memory (VRAM)	48GB
CPU	Intel(R) Xeon(R) w5-3435X
System RAM	256GB
Software Configurations	
OS	Ubuntu 22.04.5 LTS
NVIDIA Driver	535.247.01
CUDA	12.2
Python	3.8.19
PyTorch	2.4.1
Torchvision/Torchaudio	0.19.1/2.4.1
Transformers	4.31.0

$T = 10$ and batch size is 1. Additionally, we performed size transformations in both the spatial and frequency domains to enhance the diversity of adversarial examples. In the spatial domain, for each iteration, the images were processed using bicubic interpolation and the SGA [4] scale transformation sizes, adjusting the images to five different scales: $S = \{0.50, 0.75, 1.00, 1.25, 1.50\}$. In the frequency domain, the images from each iteration are transformed into the frequency domain using 2D-DCT to obtain the spectrogram. The spectrogram is multiplied by $M_\psi \in (1 - 0.5, 1 + 0.5)$. Finally, the example with the frequency-domain input scale transformation is obtained through the IDCT. All experiments are conducted on an NVIDIA RTX A6000 GPU. To facilitate reproducibility, we document the environment employed across all experiments, including hardware and software settings; details are given in Table II.

4) Evaluation Metric: The attack success rate (ASR%) is the primary metric used to evaluate the attack capability of adversarial examples on VLP downstream tasks. A higher ASR value indicates stronger attack capabilities of the adversarial examples. For the image-text retrieval task, we use recall to evaluate the performance of image-text retrieval models. Recall measures the proportion of relevant texts retrieved out of all relevant examples. In the paper, R@1, R@5, and R@10 indicate whether the correct matching examples are included within the Top 1, Top 5, and Top 10 search results, respectively. For the caption generation task, the evaluation metrics primarily selected include BLEU [51], CIDEr [52], METEOR [53], ROUGE [54], and SPICE [55]. Specifically, the BLEU score is an evaluation method based on n-gram matching, which requires reference texts as standards. It calculates the similarity between the generated text and the reference text. The higher the BLEU score, the closer the generated text is to the reference text. METEOR evaluates the similarity between the generated text and the reference text based on words and phrases. ROUGE assesses the results based on recall, calculating similarity by comparing the number of matching n-grams between the generated text and the reference text. SPICE evaluates the semantic consistency between the generated description and the reference description. The SPICE score ranges from 0 to 1, with higher scores indicating greater semantic consistency.

5) Baselines: To fully demonstrate the effectiveness of our attack method, we compare the proposed I&CA attack with the latest three multimodal VLP-based attack methods.

- Co-attack. This approach focuses on generating adversarial perturbations by optimizing the overall difference in fused feature probability distributions in a white-box setting.
- SGA. This approach focuses on increasing the diversity of adversarial examples during the optimization process and emphasizes generating transferable adversarial samples by enhancing the dissimilarity between image-text pairs at the ensemble level.
- DRA. This method is based on the SGA framework and focuses on the diversification of the intersection region of adversarial trajectories, emphasizing the diversity of both the original images and the images generated during the optimization process.

B. Experimental Evaluation

For cross-model evaluation, we used the ITR task as a representative task and conducted experiments on the Flickr30K and MSCOCO datasets. We selected ALBEF and TCL as surrogate models, and the performance results are presented in Tab. I.

Our experimental results demonstrate that I&CA achieves a high ASR (%) on both the Flickr30K and MSCOCO datasets. Under the white-box attack setting, despite the strong performance of existing methods, our approach still achieves an ASR (%) of over 99.9%, further pushing the upper bound of white-box attack performance. However, given the maturity of current white-box attack methods, we focus on the transferability of adversarial examples in our attack framework under black-box settings.

In the black-box setting, I&CA has demonstrated a noticeable performance advantage over the DRA method. Specifically, when using ALBEF as the source model and transferring the generated adversarial examples to TCL, which employs the same single-stream architecture, our method (I&CA) achieved an average ASR (%) improvement of 11.68% over the DRA method in both TR and IR tasks. Furthermore, to substantiate the cross-model transferability of our attack method, we transferred the adversarial examples to dual-stream architecture models, such as CLIP_{vit} and CLIP_{cnn}. Although the transferability was somewhat reduced when moving to different model architectures, our method still outperforms DRA by an average of 5.94%. Similarly, when TCL served as the source model, our attack framework achieved an average improvement of 3.68% in TR and 3.13% in IR tasks compared to DRA. These results indicate that our attack framework has a significant advantage in transferability across different model architectures. We attribute this outcome to the effectiveness of our attack framework in guiding the divergence of individual modality features while suppressing interactions of common features, thereby enhancing the cross-model transferability of adversarial examples.

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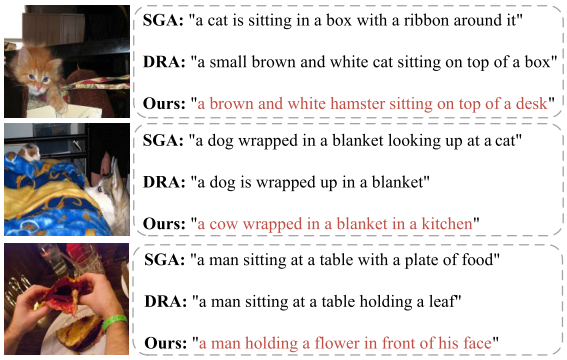

Fig. 4. Visualization of Image Captioning. ALBEF is used as a surrogate model. Transfer adversarial examples are generated for the ITR to the BLIP model, which is used to caption the adversarial images.

Fig. 5. Visualization of Visual Grounding. Using ALBEF as the surrogate model, we evaluate cross-task transferability by applying adversarial examples from ITR to visual grounding, where the model localizes regions in adversarial images based on textual captions.

TABLE III

CROSS-TASK TRANSFERABILITY. WE USE ALBEF TO GENERATE ADVERSARIAL EXAMPLES FOR ATTACKING THE VG TASK ON THE REFCOCO+ AND THE IC TASK ON THE MSCOCO. THE BASELINE REPRESENTS THE PERFORMANCE OF EACH TASK WITHOUT ANY ATTACK, WITH LOWER VALUES INDICATING BETTER ATTACK EFFECTIVENESS

Method	ITR→IC					ITR→VG		
	B@4	M	R-L	Cr	S	Val	TextA	TextB
Baseline	39.70	31.00	60.00	133.3	23.80	58.44	65.90	46.24
SGA	27.60	24.30	50.80	89.90	17.40	49.84	56.79	40.56
DRA	26.80	23.80	50.00	86.10	17.00	49.72	55.95	40.56
Ours	**22.00**	**20.90**	**45.90**	**69.30**	**14.20**	**46.92**	**54.29**	**37.59**

In addition to evaluating the transferability of I&CA, we further evaluated its transferability across different tasks. Specifically, we generated adversarial examples based on the ITR task and evaluated their impact on visual grounding and image captioning tasks. In this process, we used ALBEF as the surrogate model in all cases. For the VG task, we assessed the cross-task transferability of the generated adversarial examples on the ALBEF model. For the IC task, we evaluated the cross-model and cross-task transferability of these examples on the BLIP [56] model. As shown in Tab. III (B@4, M, R L, Cr, and S representing BLEU-4, Meteor, Rouge L, CIDEr, and SPICE, respectively). The visual results are presented in Figs. 4 and 5, which demonstrate that our framework successfully influenced both visual grounding and image captioning tasks. Furthermore, the adversarial examples generated under our framework significantly outperformed DRA in terms of cross-task performance. Specifically, when transferred to the IC task and VG task, our method achieved an average improvement of 5.7% and 2.48% compared to DRA. Therefore, our framework not only excels in cross-model transferability but also exhibits outstanding performance in cross-task transferability.

TABLE IV

OUR ATTACK FRAMEWORK ON THE ITR TASKS OF THE FLICKR30K DATASET, WITH ASR (%) UNDER DIFFERENT STRATEGY COMBINATIONS. THE COMPLETE SCHEME SHOWS THE BEST PERFORMANCE IN ALL BLACK-BOX SETTINGS

$\mathcal{L}_{Ind}$	$\mathcal{L}_{com}$	$T_{Aug.}$	$I_{Aug.}$	TCL		$CLIP_{vit}$		$CLIP_{cnn}$	
				TR	IR	TR	IR	TR	IR
✓	✓	✓	✓	**94.61**	**93.98**	**34.50**	**47.37**	**38.77**	**51.54**
✓	✓	✓	✗	87.55	87.06	34.62	47.33	38.51	51.66
✓	✓	✗	✗	83.80	85.19	34.54	48.20	38.10	50.87
✓	✗	✗	✗	80.30	80.71	28.36	40.68	33.24	43.96

C. Ablation Study

1) Strategies: To investigate the impact of the different strategies on our proposed method, we conducted ablation experiments on the ITR task using ALBEF as the surrogate model. We evaluated its transferability to three other VLP models. Tab. IV presents the ASR (%) performance under different strategy combinations, showing that the complete scheme performs best in a black-box setting. The improvement in attack performance brought by the common loss is approximately 5.62% on average. Regarding image augmentation, we applied scale transformations in the spatial domain to capture richer modal features and transformed the images into the frequency domain for scale changes. Fig. 7 displays the scale variations in different domains and the ASR (%) when both domains are combined. The results indicate that performing scale transformations in both the spatial and frequency domains leads to a higher ASR (%).

2) Weight Parameters γ and β: We evaluated the attack performance under different weights. Specifically, to present the attack effect more intuitively, we averaged R@1, R@5, and R@10 as the evaluation metrics. As shown in Fig. 8 (a), (b), (c), and (d). The ASRs (%) of adversarial examples are displayed when the parameters change with weights of 0.1, 1, 10, and 100. The results indicate that the attack performance is optimal when $\gamma = 10$ and $\beta = 1$.

3) Scaling Parameter ρ: In the frequency-domain scaling transformation, we control the degree of transformation through $M_\psi \in (1-\rho, 1+\rho)$, where $\rho \in (0, 1)$. To investigate the impact of the scaling factor matrix on attack performance, we conducted independent experiments on the value of ρ.

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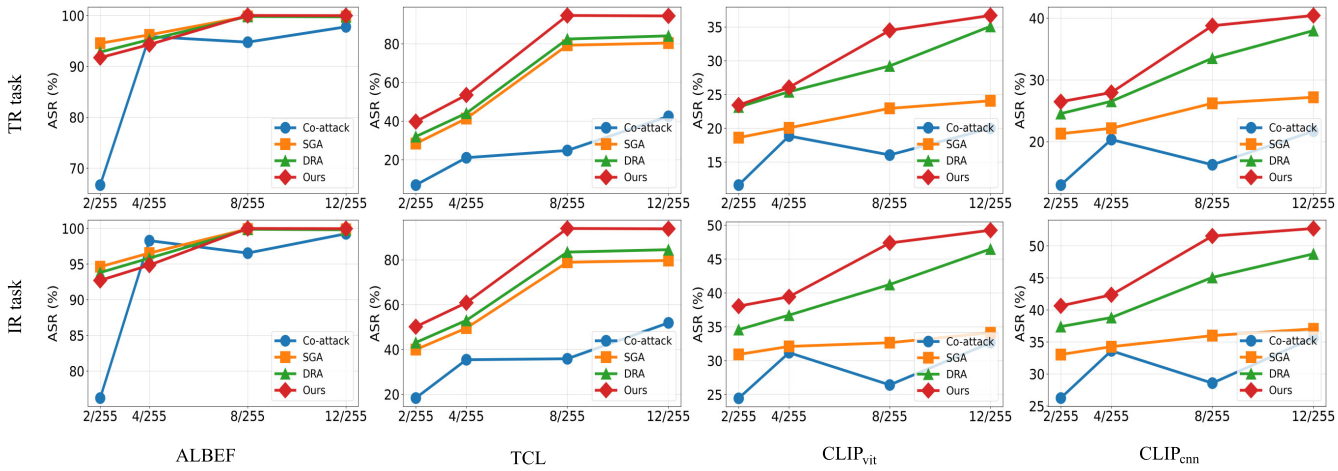

Fig. 6. As a surrogate model, ALBEF was used to evaluate the performance of our attack framework in TR and IR tasks under different perturbation settings. Subsequently, it was transferred to the TCL, CLIP$_{vit}$, and CLIP$_{cnn}$ models for further evaluation.

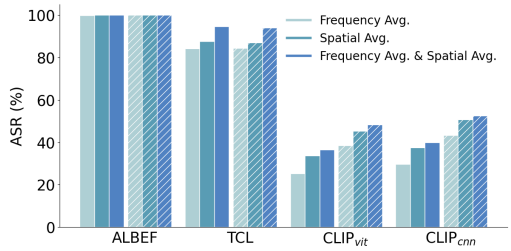

Fig. 7. ASRs (%) under the spatial domain, frequency domain scale variations, and mixed scale transformations using ALBEF as the surrogate model. The 'diagonal stripe' represents the IR task, and the rest are for the TR task.

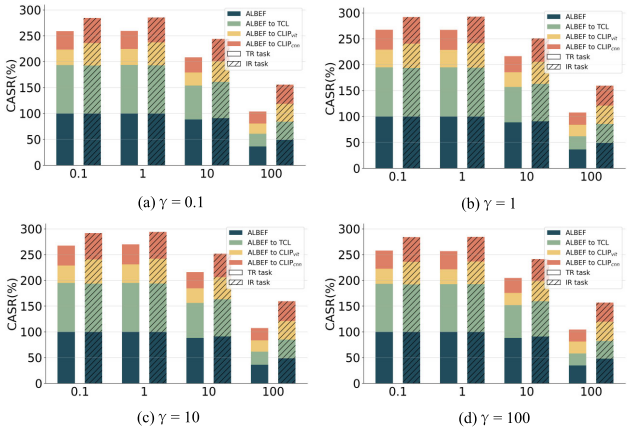

Fig. 8. ASRs (%) under the corresponding transformations of β using ALBEF as the surrogate model. The weight parameter γ is set to 0.1, 1, 10, and 100. "CASR (%)" stands for Cumulative Attack Success Rate.

Specifically, using the Flickr30K dataset, we performed experiments on the ALBEF model for the ITR task, testing the attack effects with scaling factor values of 0.1, 0.3, 0.5, 0.7, and 0.9. The experimental results are shown in Tab. V. According to our experimental results, the value of the scaling factor ρ does not have a significant impact on the outcomes. Therefore, throughout our experimental setup, we chose $\rho = 0.5$ as the default value.

TABLE V

THE IMPACT OF THE VALUE OF ρ ON THE EXPERIMENTAL RESULTS IS PRESENTED HERE, SHOWING ONLY THE MEAN VALUES OF R@1, R@5, R@10

ρ	ALBEF		TCL		CLIP$_{vit}$		CLIP$_{cnn}$	
	TR	IR	TR	IR	TR	IR	TR	IR
0.1	100.00	99.99	88.50	88.64	33.79	46.63	37.74	50.79
0.3	100.00	100.00	92.45	92.05	33.48	47.01	37.78	51.06
0.5	100.00	99.99	94.61	93.98	34.50	47.37	38.77	51.54
0.7	99.93	100.00	95.49	93.98	34.08	47.76	38.89	51.17
0.9	99.87	99.96	93.46	93.40	33.90	47.73	38.89	51.13

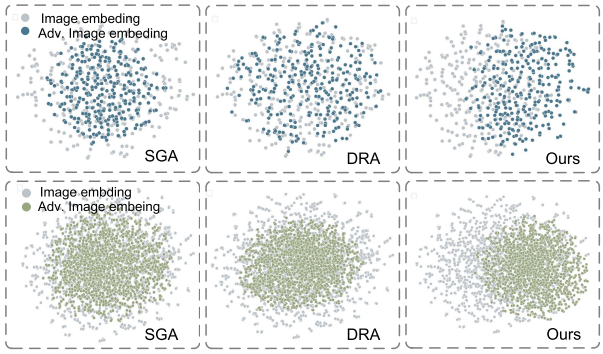

Fig. 9. T-SNE visualization of the embedded features for adversarial example pairs.

4) Perturbation Size δ_v: We evaluates the ASR (%) under different perturbation settings, including 2/255, 4/255, 8/255, and 12/255. In the white-box setting, our attack framework demonstrates a gradual improvement in performance as the perturbation size increases, eventually stabilizing. Although the attack performance in certain white-box scenarios is slightly lower than that of Co-Attack, the transferability in the black-box setting is significantly higher than that of current studies.

5) Embedding Features: Embedding Features.We randomly selected 300 pairs of adversarial samples and used the T-SNE

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TABLE VI
COMPARISON OF TIME COSTS. HERE, t_I REPRESENTS THE TIME CONSUMED TO GENERATE AN IMAGE, AND t_T REPRESENTS THE TIME CONSUMED TO GENERATE TEXT

attack	ALBEF		TCL	
	$t_I + t_T$	t_I	$t_I + t_T$	t_I
SGA	2.12	1.64	2.01	1.62
DRA	4.85	4.25	4.53	4.31
Ours	3.01	2.53	2.84	2.28

TABLE VII
THE ATTACK SUCCESS RATES OF DIFFERENT ATTACK METHODS WHEN TRANSFERRED TO LVLMS

Method/Model	Pandagpt_7B	BLIP2	MiniGPT-4_7B
Co-attack	17.58	2.86	15.24
SGA	18.78	7.28	16.02
DRA	19.76	7.96	16.44
Ours	**22.64**	**11.44**	**18.30**

method to visualize the embedded features of the generated adversarial samples. The results are shown in Fig. 9. Through the T-SNE feature distribution visualization, the adversarial samples generated by I&CA exhibit greater feature space divergence and dispersion compared to SGA and DRA. The features generated by SGA and DRA are typically clustered, and the perturbation of individual features alone causes the adversarial samples to fit the specific decision boundary of the surrogate model, limiting their transferability to other models. In contrast, I&CA explicitly models both common and individual features, and further perturbs the model's learning of shared information based on individual features. This causes the generated adversarial sample features to be significantly more dispersed, pushing the embeddings away from the clean feature manifold, thereby inducing a generalized semantic shift, which ensures better transferability.

D. Time Cost Comparison

Compare the time required to generate each adversarial example using different attack methods in Tab. VI. Compared to SGA, I&CA takes slightly longer to generate adversarial examples. However, based on previous experimental results, the attack performance of I&CA is significantly higher than that of SGA. Compared to DRA, our I&CA method outperforms the DRA attack in both time efficiency and attack performance.

V. TRANSFERABILITY ON LVLMS

In recent years, large visual-language models (LVLMs) have made significant progress in multimodal tasks. To further explore the impact of adversarial examples generated by VLP models on LVLMs, we conducted a series of extended experiments.

We used the Flickr30K dataset (includes 1000 images, each annotated with five captions) and selected the ALBEF model as a surrogate model to generate adversarial examples under $\epsilon_t = 1$ and $\epsilon_v = 8/255$. Subsequently, we fed these adversarial examples into various LVLMs, including Pandagpt [57], BLIP2 [58], and MiniGPT-4 [59], for evaluation. Due to the differences in task design between VLP models and LVLMs, we redesigned the ITR task on LVLMs. Specifically, we input the adversarial images v' generated by the ALBEF model along with the original text $\boldsymbol{t} = \{t_1, t_2, \ldots, t_m\}$ into the LVLMs and modified the text prompt to "Does this image describe the 't_i'? Answer only with 'yes' or 'no'." By comparing the responses of LVLMs on v and v', we considered the attack successful if the answers were inconsistent. The relevant experimental results are detailed in Tab. VII. Additionally, we conducted cross-task experiments on LVLMs. Specifically, the adversarial examples v' generated by ALBEF for the ITR task were transferred to BLIP2 and InstructBLIP [60] models for the caption generation task. The experimental results are shown in Tab. VIII. The experimental results indicate that, although the I&CA attack was designed for VLP models, it still interfered with the judgment ability of LVLMs to some extent, both in cross-model and cross-task scenarios. Moreover, the attack performance of I&CA on LVLMs was superior to the baseline methods, further highlighting the necessity of exploring the robustness of LVLMs in the future.

VI. ANALYZE FURTHER

Through the experiments conducted above, it was found that the I&CA attack method outperforms current other attack methods. In this section, we will further analyze why I&CA achieves better performance and why the current methods fail to effectively capture the relevant features.

Why does I&CA achieve better performance?

- Different models exhibit overlapping attention regions to common features during feature fusion. To verify this, we selected 300 image-text pairs (with five descriptions for each image) and compared the ALBEF, TCL, and BLIP models (where ALBEF and TCL share the same architecture but have different training objectives, while ALBEF and BLIP belong to different model architectures). We focused on analyzing the differences in attention to interaction information across these models when processing image-text pairs. Specifically, we extracted the attention scores from each layer of the models and averaged the scores across multiple layers to obtain an attention score matrix. Then, we analyzed the spatial distribution differences of the top-k values in the attention score matrices between different models.Specifically, we quantify the overlap between attention regions of different models using the Intersection over Union (IoU), calculated as follows:
$$\mathrm{IoU}(A_1, A_2) = \frac{|A_1 \cap A_2|}{|A_1 \cup A_2|}$$
Where A_1 and A_2 represent the attention score matrices from different models. Since ALBEF and BLIP belong to different model architectures, their attention score matrices have different dimensions. Therefore, we averaged the scores along the text dimension and compared them, with the results shown in Fig. 10. The visualization results are shown in Fig. 11. We found that, regardless of whether

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TABLE VIII

THE ADVERSARIAL EXAMPLES GENERATED BY ALBEF FOR THE ITR TASK ARE TRANSFERRED TO LVLMS FOR EVALUATION ON THE IMAGE CAPTIONING TASK, WHERE LOWER VALUES INDICATE BETTER ATTACK PERFORMANCE

model	attack	Flickr30k					MSCOCO				
		B@4	M	R-L	Cr	S	B@4	M	R-L	Cr	S
BLIP2	baseline	41.45	27.20	57.96	100.95	13.32	37.13	28.00	57.99	123.80	21.98
	Co-attack	25.23	20.57	48.29	60.42	14.08	32.92	25.58	55.18	108.81	19.66
	SGA	24.16	19.47	46.98	55.77	13.27	31.78	25.02	54.36	104.93	19.06
	DRA	23.60	19.61	46.90	55.58	13.15	31.68	24.94	54.24	104.53	19.02
	Ours	**21.80**	**18.32**	**45.78**	**51.60**	**12.40**	**31.09**	**24.63**	**53.85**	**102.65**	**18.60**
instructBLIP	baseline	47.57	30.98	62.49	122.17	15.64	39.36	31.64	59.41	141.85	24.93
	Co-attack	25.89	22.23	48.51	67.52	15.33	36.80	28.90	57.61	123.93	22.31
	SGA	24.42	21.00	46.91	61.72	14.35	35.00	27.93	56.34	117.20	21.21
	DRA	23.41	20.72	46.65	60.14	14.29	34.62	27.81	56.12	116.31	21.21
	Ours	**21.71**	**19.70**	**45.73**	**59.00**	**13.16**	**33.12**	**26.60**	**54.70**	**114.12**	**20.55**

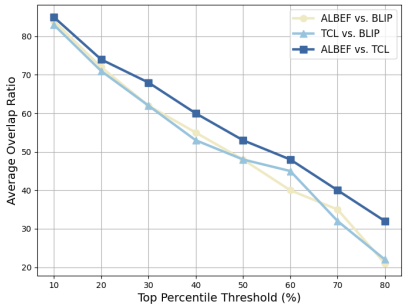

Fig. 10. The spatial distribution of attention to image-text fusion information across different models reflects their differences in attention to common features during modality interaction.

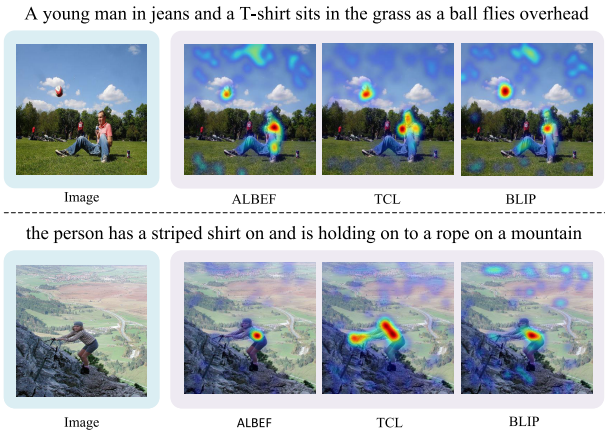

Fig. 11. Visualization of how different models (ALBEF, TCL, and BLIP) respond to common features during the feature fusion process.

the models have the same architecture but different training objectives or different architectures, there is spatial overlap in their learning of key information in modality interaction. If we define the top 30% of attention scores as the information most focused on by the model, we found that these pieces of information have at least 60% spatial overlap.

- The rationale behind the design of individual loss and common loss. Since different models exhibit overlapping attention regions to common modality features, suppressing the interaction of common features can improve transferability. Moreover, the expression of common features depends on the learning of individual features from each modality in the previous stage. Therefore, individual loss is used to disrupt the learning process from the previous stage.
- Dual-modal augmentation effectively alleviates the overfitting problem. Since the perturbation update of one modality depends on another modality, this may lead to the perturbation being overly fitted to the learned biases of that modality's model, thus limiting transferability. In the experiments in the next section, we verified the overfitting problem present in current attack methods. Finally, through attribution analysis in the ablation study, as shown in Tab. IV, we found that dual-modal augmentation based on common loss and individual loss led to an average increase of 3.34% in attack success rate, indicating that dual-modal augmentation effectively mitigates the overfitting problem.

Why the current method fails to effectively capture relevant features?

- Different models exhibit overlapping attention regions to common features during feature fusion. However, both SGA and DRA focus solely on disrupting the alignment of individual modality features, neglecting the common feature information in the modality fusion process.
- The perturbation update of one modality relying solely on the feature information of another modality leads to severe overfitting. Both DRA and SGA generate adversarial examples in three stages:

$$v \xrightarrow{\text{Stage 1}} t' \xrightarrow{\text{Stage 2}} v' \xrightarrow{\text{Stage 3}} t''$$

In the entire process, the generation of the adversarial text t''relies on the information from the image modality (i.e., the individualized features of the image) multiple times,

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TABLE IX

COMPARISON OF OVERFITTING EXPERIMENTS. HERE, SGA_3 AND SGA_2 DENOTE THE THREE-STAGE AND TWO-STAGE VARIANTS OF THE SGA ATTACK, RESPECTIVELY, AND THE CELLS WITH GRAY BACKGROUNDS CORRESPOND TO THE WHITE-BOX ATTACK SETTING

Flickr30K (Image-Text Retrieval)													
Model	Method	ALBEF			TCL			$CLIP_{vit}$			$CLIP_{cnn}$		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
ALBEF	SGA_3	**100.00**	99.70	99.70	87.57	77.69	72.44	36.93	18.90	13.01	41.00	21.88	15.76
	SGA_2	99.97	**99.80**	**99.80**	**89.20**	**79.41**	**74.12**	**40.17**	**21.97**	**15.63**	**45.04**	**25.78**	**18.89**
	DRA_3	99.79	99.80	99.80	90.62	81.11	75.55	45.40	24.51	17.78	50.45	29.07	21.01
	DRA_2	**99.79**	99.70	99.70	**91.25**	**82.81**	**77.96**	**47.15**	**26.84**	**19.76**	**52.90**	**31.14**	**22.41**
TCL	SGA_3	93.12	86.97	84.00	100.00	100.00	100.00	41.96	24.40	16.77	50.70	29.81	20.29
	SGA_2	**93.33**	**88.18**	**85.30**	**100.00**	**99.70**	**99.70**	**43.15**	**25.67**	**17.97**	**51.60**	**30.12**	**20.97**
	DRA_3	94.68	90.48	87.90	100.00	100.00	99.90	46.85	26.34	18.50	51.11	29.89	21.43
	DRA_2	**95.19**	**91.58**	**88.10**	**100.00**	**100.00**	**99.97**	**47.76**	**26.78**	**18.99**	**52.09**	**31.79**	**22.09**
Flickr30K (Text-Image Retrieval)													
Model	Method	ALBEF			TCL			$CLIP_{vit}$			$CLIP_{cnn}$		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
ALBEF	SGA_3	99.93	99.92	99.90	87.64	77.47	71.61	46.97	28.75	22.20	50.43	32.78	24.76
	SGA_2	**99.98**	**99.96**	**99.88**	**89.69**	**78.03**	**73.46**	**48.37**	**31.42**	**24.69**	**55.28**	**34.72**	**28.17**
	DRA_3	99.93	99.88	99.84	91.00	81.99	77.34	56.48	38.05	29.18	59.42	41.99	33.84
	DRA_2	**100.00**	**99.98**	**99.96**	**93.23**	**83.69**	**80.55**	**58.11**	**39.74**	**30.20**	**61.34**	**43.05**	**35.11**
TCL	SGA_3	92.52	86.26	82.39	100.00	99.98	99.98	49.90	32.96	25.67	55.99	37.84	29.07
	SGA_2	**92.95**	**88.84**	**84.84**	**100.00**	**99.98**	**99.98**	**52.95**	**35.96**	**57.52**	**58.33**	**40.46**	**31.34**
	DRA_3	94.95	90.20	87.63	100.00	99.98	99.96	56.84	38.95	30.99	62.11	43.76	35.41
	DRA_2	**95.23**	**90.48**	**87.93**	**100.00**	**99.98**	**99.96**	**56.98**	**39.12**	**32.23**	**62.24**	**44.69**	**35.98**

which leads to the text perturbation overfitting to the learning preferences of the image modality. To this end, we conducted comparative experiments on the Flickr30K dataset by directly removing the third stage from the SGA and DRA attack methods. The experimental results are shown in the Tab. IX. Based on the results, we found that by simply removing the third stage, without making any other modifications, the attack success rates of SGA and DRA increased by an average of 2.21% and 1.64%, respectively. Compared with SGA, the improvement of DRA is smaller because SGA itself already applies diverse processing to images, which, to some extent, alleviates the risk of perturbation optimization overfitting to the modality's learning preferences, but it ignores the semantic diversity of the text modality.

VII. CONCLUSION

This study primarily addresses the security risks inherent in Vision–Language Pretrained (VLP) models and, in view of their vulnerability to adversarial attacks, introduces a novel and efficient attack framework named I&CA. The framework innovatively takes into account both modality individual features and cross-modal common features. By inducing divergence among individual features while suppressing the expression of common features, I&CA substantially enhances the generalization and transferability of the crafted adversarial examples.

Extensive experiments show that, across multiple mainstream VLP models and diverse tasks, adversarial examples generated by **I&CA** achieve markedly better transfer performance than those produced by the current state-of-the-art DRA attack, with an average improvement of 6.15%. This gain not only validates the effectiveness of our framework design, but also underscores the importance of jointly optimizing individual and common features when crafting adversarial perturbations. Notably, we further demonstrate that adversarial examples tailored for VLP models can also successfully mislead LVLMs, and that **I&CA** exhibits superior transferability across both models and tasks. These findings reveal widespread security vulnerabilities in contemporary vision–language models and highlight the necessity of developing more robust training mechanisms. We hope this work offers new perspectives for the security of multimodal models, stimulates research into effective defense techniques and training strategies, and ultimately enhances the robustness and safety of both VLP models and LVLMs in real-world applications.

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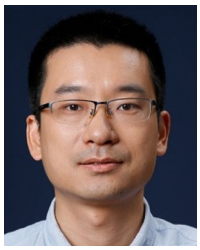

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